Object-Centric Counterfactual Critics for Robust Grasp Candidate Selection

***Abstract*—A learned critic that reranks grasp candidates before execution is only as good as the causal validity of its training labels and the fairness of the evaluation that measures it. We show both can silently break, and how to catch it. A pre-execution grasp critic trained on simulated MuJoCo outcomes initially appeared to beat a geometric heuristic baseline by +15.6 percentage points – a result that collapsed under scrutiny: the evaluation harness encoded the compared method into its random seed, so paired trials never shared a scene, and the success label counted mere gripper contact regardless of whether the object was lifted. Under a rebuilt, shared-candidate-pool paired evaluation, the original critic's success probability carries no real signal (AUROC= 0.4996 – chance level). We then train an object-relative counterfactual critic – pose relative to the target object, point-cloud statistics, and object identity, trained with a within-scene Bradley–Terry-style pairwise loss on causally-admissible features only – and evaluate it on two disjoint, held-out scene batches. It significantly outperforms the geometric baseline on both: +15.6pp (p=0.00258, n=90) on an independent development-test batch, and +14.0pp (p=3.24e-4, n=150) on a frozen confirmatory batch never inspected before evaluation. A complementary offline decomposition explains where and why the critic works. We report negative results with equal rigor: an uncertainty-based risk gate adds no measurable benefit, and the pairwise loss term's independent contribution is quantifiably unresolvable at the current data scale. We release a causal-validity audit tool, a paired-evaluation harness, and a real-hardware validation protocol for future work.**

I. Introduction

Pre-execution grasp candidate scoring – training a model to rank a pool of not-yet-executed grasp poses so the best one can be selected before any physical or physically-simulated commitment – is a common pattern across analytic scorers [1], learned discriminators [4], and geometric rerankers, including this project's own LGGSN (companion T-RO work). Its correctness rests on two assumptions that are easy to state and easy to violate silently: every input feature must be computable *before* the chosen candidate executes (causal validity), and comparing two scoring methods requires evaluating them against the same candidate pool under the same conditions (paired evaluation). Neither violation is visible from an aggregate accuracy or success-rate number alone – both look like a real result until traced to source.

We hit both violations in the same pipeline. A pre-execution grasp critic ("world model"), trained on simulated MuJoCo outcomes, was originally reported to beat a geometric heuristic baseline by +15.6 percentage points. Auditing this claim before building on it – applying this project's own PRE_EXECUTION-admissibility criterion and provenance registry (companion T-RO work) to a new pipeline – surfaced two compounding defects, neither itself a feature-provenance violation: the evaluation harness's random seed encoded which method was under test, so "geometry" and "critic" trials never executed against the same scene or candidate pool despite the paired statistics assuming they did; and the success label counted transient post-close gripper contact, checked before any lift, regardless of whether the grasp actually succeeded. Re-evaluated under a from-scratch, causally-admissible, genuinely paired harness, the original checkpoint's apparent signal is AUROC= 0.4996 against 1,500 real per-candidate outcomes – exactly chance.

Rather than discard the direction, we rebuilt it. An object-relative counterfactual critic – pose features expressed relative to the target object, point-cloud statistics, and object identity, every field registered PRE_EXECUTION-admissible *before* training, not after – significantly outperforms the geometric baseline on two independent, disjoint, held-out scene batches, one of them a frozen confirmatory batch never inspected before its pre-registered gate was evaluated. We report this result with the same statistical discipline that caught the original claim's invalidity, and we report, with equal weight, what this investigation could not establish: an uncertainty-based risk gate with no measurable benefit, a pairwise-loss ablation quantifiably not resolvable at the current evaluation scale (section V-C), and a small-data imitation-learning integration that has not yet demonstrated closed-loop robustness.

This paper is organized around that arc – audit, diagnose, rebuild, re-verify – mirroring this project's companion T-RO investigation into the same class of failure in a different pipeline. We do not claim the specific defects found here are universal; we claim the failure mode – an aggregate metric that cannot distinguish a causally valid, fairly evaluated result from an invalid one – is a structural risk for any pipeline in this family, and that catching it requires checking an evaluation's construction, not only its output.

Relation to companion T-RO work. The PRE_EXECUTION-admissibility criterion and its audit tooling are this project's own prior formalization, introduced and validated in a companion T-RO submission on a different, retrospective pipeline (cross-embodiment LGGSN reranking). This paper's contribution is not the criterion itself, but its *prospective* application to a second, independent pipeline, plus a new critic architecture built causally-admissible from the start and validated with two disjoint held-out test batches – see section II for how the two papers are positioned to avoid overlapping claims.

II. Related Work

Learned grasp-candidate scoring and reranking. Dex-Net 2.0 [1] and GraspNet-1Billion [2] learn grasp-quality predictors from large annotated/simulated datasets assuming full scene geometry at query time; 6-DOF GraspNet [3] extends this to a variational proposal generator. None of the three states an input/label admissibility distinction as a formal, checkable criterion. GraspGen [4] is the closest prior system to satisfy the PRE_EXECUTION-admissibility criterion we apply here by construction: its discriminator's inference-time inputs are a point-cloud encoding and candidate pose only, with execution-derived signal confined to training labels. As in this project's companion T-RO work, we treat GraspGen as a confirmed-compliant reference case, not a counterexample – external validation of the criterion, not prior art for it.

Evaluation-protocol failures in offline policy/critic assessment. Data leakage and unfair offline comparison are recognized general risks in sequential-decision-making evaluation. Concurrent work on offline policy evaluation for manipulation, "Offline Policy Evaluation for Manipulation Policies via Discounted Liveness Formulation" [5], diagnoses a *different* integrity problem from ours: finite-horizon episodes bias value estimates under classical TD/Monte Carlo evaluation (truncation bias), which they address with a liveness-based Bellman operator, evaluated on VLA and diffusion-policy manipulation tasks plus cloth folding. This is adjacent evidence that offline/simulated manipulation-policy evaluation is error-prone in more than one structurally distinct way – not directly overlapping prior art for the seed-coupling and success-criterion defects section IV diagnoses, and should not be read as a leakage paper. Independently, and closer in spirit to our diagnosis, "What Are We Actually Benchmarking in Robot Manipulation?" [6] audits LIBERO and CALVIN and finds only 19.8% of LIBERO's reported advances are statistically significant once evaluation noise is accounted for, alongside shortcut-solvable benchmark items and train/test-distribution proximity effects that inflate apparent generalization. Their specific failure modes (shortcut solvability, creeping overfitting, data-source dependence) are distinct from the seed-coupling and success-criterion defects section IV diagnoses in our own pipeline, but the broader concern is the same and, as of mid-2026, actively converging from multiple independent directions: a robot-manipulation benchmark score is not self-certifying, and treating one as evidence of general capability without auditing how it was produced is an increasingly recognized, not a niche, risk. We position this paper as one concrete, fully-traced instance of that broader concern, not as its sole discovery.

Risk-aware and uncertainty-gated action selection in VLA policies. ReconVLA [7] applies conformal prediction directly to a pretrained VLA policy's action tokens, yielding calibrated uncertainty estimates that correlate with execution quality, and extends the same idea to state-space outlier detection as a failure-anticipation mechanism. SafeVLA [8] frames safety alignment as a constrained Markov decision process, optimizing against elicited unsafe behaviors and reporting an 83.58% reduction in cumulative safety-violation cost versus prior methods. Both report a positive effect from their respective uncertainty/safety mechanism. Our own ensemble-disagreement risk gate – calibrated on held-out data and evaluated with the same statistical rigor as our positive result – shows no measurable benefit over the ungated critic (coverage 98.7%, section VI). We do not read this as evidence against uncertainty-gating in general: ReconVLA's per-token conformal calibration and SafeVLA's constrained-optimization framing are both structurally different from a post-hoc ensemble-disagreement filter applied to a pre-execution candidate scorer. Concurrent work directly asking when uncertainty-gating helps, "Confidence-Gated Robot Autonomy" [9], identifies a dataset-dependent *competence regime*: below it, uncertainty rankings are weak and unstable regardless of which uncertainty source (softmax, MC dropout, ensemble) is used; above it, simple proxies suffice and the gating threshold matters more than the method. Our critic's $\sim 50\%$ pooled success rate (section V) may simply not yet be in the regime their analysis identifies as necessary for a gate to add value – we flag this as the most likely explanation for our null result rather than leaving it unexplained, without claiming to have tested for the regime directly. We report our own result as a specific negative finding for the specific gate design tested, with the same discipline as our positive finding, not omitted or hedged into ambiguity.

Sim-to-real transfer for grasp success prediction. Out of scope for this paper's evidence base; motivates section VII's protocol as a stated future-work step, not a claim made here.

III. Method

A. *Causal-validity criterion (reused, not reinvented)*

We reuse Definitions 1–3 and the audit algorithm from this project's companion T-RO formalization at the level of generality needed here: a scene induces a candidate pool, each candidate carries generator-time attributes, and a feature is PRE_EXECUTION-admissible only if it is computable from information available before the candidate it describes has executed – with training labels themselves exempt (they are necessarily execution-derived by definition, and only input features are constrained). This paper's contribution here is *application and extension* (registering a new pipeline's fields, section III-B), not the criterion itself.

B. *Object-relative counterfactual critic*

The candidate pose is expressed relative to the detected/simulated object position, with sin/cos yaw, gripper opening, 9-dimensional point-cloud statistics, and an object-identity one-hot (3 classes: CrackerBox, MustardBottle, PowerDrill). All fields are PRE_EXECUTION-admissible per the section III-A criterion – a checked property, registered in the project's feature-provenance registry, not an assumption. The architecture is a 2-hidden-layer MLP (64/64, SiLU, dropout 0.05), trained as a 5-seed ensemble. The loss is binary cross-entropy on the per-candidate ground-truth outcome, plus, for the `object_counter factual` variant, a within-scene Bradley–Terry/BPR-style pairwise term comparing successful vs. failed candidates from the *same* scene – the "counterfactual" framing: what would have happened had a different candidate in the same, already-observed scene been chosen. This is a same-scene, within-episode alternative used as a *training signal* for the critic, not the post-hoc explanation or formal potential-outcomes sense of "counterfactual" common in the 2025–2026 causal-ML and explainability literature – the pairwise comparison never requires estimating an outcome under an unobserved intervention, only comparing two candidates both already scored against the same, already-observed scene. The training/validation split is scene-grouped (never splitting a scene's candidates across train/val), stratified per object, and reproducible per seed.

C. Paired candidate-selection evaluation harness

One candidate pool (object placement plus K sampled grasp poses) is built before any method is chosen; every compared method executes against a *fresh, identically replaced* copy of the same scene, sharing the same pool. This directly fixes the predecessor pipeline's seed-coupling defect (section IV-A). The production execution primitive is the same bilaterally-gated, weld-based physics grasp routine used elsewhere in this project's simulation stack, not a bespoke, more lenient reimplementation (section IV-B shows why that distinction matters). Statistics are McNemar's exact test (paired), bootstrap confidence intervals, and per-object and pooled effects, with all gates pre-registered before any result was inspected.

IV. Diagnosing the Predecessor Pipeline

This section is itself a result, not just setup, matching this project's established house style of reporting a self-correction as a first-class finding rather than suppressing it.

A. The seed-coupling defect

The predecessor harness's random-number-generator seed was derived in part from the identity of the method under test. This is provably method-dependent by construction, and empirically confirmed directly: of 250 paired trials, zero shared even the same executed candidate position between the "geometry" and "world-model" arms – the paired design the original statistics assumed was never actually realized.

TABLE I
Paired live-executed results: geometry vs. the corrected, object-relative counterfactual critic.

Batch	n	Geometry	Critic	Δ / exact McNemar
Dev-test	90	33.3%	48.9%	+15.6pp, p=0.00258
Confirmatory	150	36.0%	50.0%	+14.0pp, p=3.24e-4

B. The success-criterion defect

The legacy success criterion was `contact or grasped or lifted`, evaluated immediately on gripper close. This criterion saturates: across 2 objects and 15 scenes, 150/150 sampled candidates "succeeded" regardless of pose. The corrected criterion instead requires bilateral contact plus a verified post-settle lift, enforced by the production grasp primitive (section III-C).

C. Consequence

The originally-reported effect (+15.6pp, "world model" vs. geometry) is unsupported by either the pairing or the underlying success labels. This is why the paper's positive result (section V) is built on a from-scratch corrected pipeline, not a patch to the original one.

V. Results

A. Stale-checkpoint gate (negative, reported first)

Matching this project's own pre-registered-gate discipline, we report the negative result before the positive one. The original critic checkpoint, re-evaluated under the corrected pipeline, fails the pre-registered gate: pooled effect -14.0pp (wrong sign vs. the required $\geq +8$pp), AUROC= 0.4996 on 1,500 real per-candidate labels (chance level). This is not a mysterious failure: section IV-B's success-criterion defect contaminated its training labels too.

B. Object-relative counterfactual critic – the positive result

Table I summarizes both evaluations. The independent development-test batch (base seed 200, n=90) shows geometry at 30/90 (33.3%) vs. the critic at 44/90 (48.9%), +15.6pp (exact McNemar p=0.00258). The frozen confirmatory batch (base seed 300, n=150, never inspected before its pre-registered gate) shows geometry at 54/150 (36.0%) vs. the critic at 75/150 (50.0%), +14.0pp (27 wins/6 losses, exact McNemar p=3.24e-4). Both batches' scene keys are disjoint from training (base seed 100) and from each other, independently verified via pairwise key-intersection checks, not merely asserted.

Per-object breakdown (live-executed, geometry / counterfactual). Development test: CrackerBox 4/30 vs. 4/30 (tie), PowerDrill 7/30 vs. 18/30, MustardBottle 19/30 vs. 22/30. Frozen confirmatory: CrackerBox 9/50 vs. 9/50 (tie, unchanged), MustardBottle 29/50 vs. 42/50, PowerDrill 16/50 vs. 24/50. The pooled gain is concentrated in PowerDrill and MustardBottle at both evaluation scales; CrackerBox is flat at both – the critic never selects a different top-1 candidate than geometry for this object, at either scale, not a near-miss that a larger sample might resolve.

Live-executed vs. offline-re-scored reporting convention. `geometry` and `object_counterfactual` were the two methods actually live-selected during data collection, so their reported numbers are live-executed paired outcomes – the real online trial, not a re-run. The BCE-only ablation (section V-C) and section V-D's multi-head decomposition were never live-selected in this collection run; their numbers are offline re-scores against the same scene's fully-swept candidate ground truth. This distinction matters for a subtler reason than convenience: MuJoCo's contact solver is not perfectly reproducible on marginal grasps. Independently verified across the confirmatory batch's 300 live-vs-reswept comparisons, exactly 2 flipped (a $\sim$0.67% marginal-grasp non-determinism rate; the development-test batch shows the same pattern, 1 flip per method out of 90) – a genuine physical-reproducibility floor on any single-run success-rate estimate in this simulator, not a script bug, and the same class of finding independently documented in a companion pipeline ($\sim$12.5% single-run instability). We report it as a stated limitation rather than treating any single execution as ground truth.

C. Ablation: is the pairwise loss load-bearing?

Object-relative BCE alone (no pairwise term) is statistically indistinguishable from the full `object_counterfactual` variant on the frozen confirmatory batch (2 wins/3 losses, p=1.0). This is not merely "n is small" – it is quantifiably not resolvable by modestly scaling up the same evaluation: exact power analysis on the observed 2-vs-3 discordant split gives essentially zero post-hoc power and a required **210 discordant pairs** for 80% power at this effect size – against only 5 observed here out of 150 evaluated pools, resolving this would need on the order of several thousand additional evaluated candidate pools, not a modest top-up. We do not attempt to close this ablation by collecting more data at the current scale, and scope this paper's central claim to avoid depending on it (section VIII): the defensible claim is "object-centric, causally-admissible scoring beats geometry," not "the within-scene pairwise loss term is what makes it work."

D. Mechanistic analysis: multi-head decomposition (offline, complementary)

Section V-B establishes *that* the corrected critic beats geometry, live-executed – the paper's primary quantitative claim. This subsection asks *why and where* it works, via a second, explicitly **offline re-scoring** experiment on the same causally-admissible features. It is complementary evidence, not a replication, and must never be pooled with or presented as confirming section V-B's live-executed numbers.

The same object-relative feature set is decomposed into four prediction heads instead of one: `bilateral_contact`, `lifted`, `success` (binary), and a 3-class `failure_type` (`success`/`no_contact`/`weld_no_lift`). Two additional classes originally specified, `contact_no_weld` and `lifted_then_dropped`, were confirmed **structurally absent** from every collected batch – the weld gate is currently identical to bilateral contact by construction, and the `fell_off` threshold never triggers within the post-grasp settle window – so the 3-class scheme is what was actually trained; the 6-class taxonomy is documented future schema, not a current claim. Training reuses the same base-100/dev-200/confirmatory-300 split chain as section V-B, with checkpoint and loss-weighting choice frozen on dev-200 only and confirmatory-300 read exactly once.

The success head alone, used for top-1 selection, beats geometry pooled +10.7pp (46.0% vs. 35.3%, McNemar p=0.0052) on the offline confirmatory-300 batch – directionally and in rough magnitude consistent with section V-B's live-executed +14.0pp, reported as separate, complementary evidence for the same claim, not pooled or averaged with it. All three heads reach AUROC 0.90–0.93 with ECE$\approx$ 0.04 (reasonably calibrated).

Interpretability payoff. The failure-type confusion matrix exposes a specific, actionable asymmetry a single-head success critic cannot show: 92.2% recall on true `no_contact`, but 32% of true successes are misclassified as `no_contact` – the model is conservative/under-confident on success, not dangerously over-confident on failure.

Limitation, load-bearing, not a footnote. `weld_no_lift` support is 100% drill-attributed (118/118); with PowerDrill excluded, training support for that class collapses to 1 example. This must be reported as a within-drill measurement, not a demonstrated cross-object failure-mode generalization capability. A genuine leave-one-object-out test was not run in this pass.

A methodological note worth keeping, not hiding. The `equal` vs. `success_weighted` loss-weighting ablation produced measurably different trained weights (confirmed by direct parameter comparison) but *identical* argmax-based top-1 accuracy at every one of 5 seeds on the small internal validation split used during training – only the larger dev-200 batch's continuous AUROC distinguished them. Coarse top-1 metrics can be an insufficiently sensitive selection criterion for this kind of ablation at small validation-set scale; this was caught by cross-checking, not silently reported as "no difference."

Neither this subsection nor section V-B licenses a risk-gate or VLA-policy improvement claim (section VI's negative results on both stand unchanged) – the AUROC/interpretability gains here are about the critic's own outputs, not about downstream gating or imitation-policy performance.

VI. Limitations

Per this study's own rule – do not report only positive results; distinguish evidence from hypothesis from refuted conclusion – we list every negative or unresolved finding this investigation produced, condensed to the subset that bears directly on this paper's central claim.

1) **Risk gate.** No measurable benefit over the ungated critic (coverage 98.7% – the gate rarely actually fires). We do not claim a gating contribution. The most likely explanation, per concurrent work on when uncertainty-gating helps at all [9] (section II), is that our critic's $\sim$50% pooled success rate is not yet in the competence regime that analysis identifies as necessary – stated as the most likely explanation, not a claim we separately tested for it.
2) **Pairwise-loss contribution** unresolved, and not cheaply resolvable at this evaluation scale (section V-C – quantified: $\sim$210 discordant pairs needed for 80% power, against 5 observed). The paper's central claim does not depend on resolving this.
3) **Simulation-only**: no real-hardware validation in this paper (section VII states the plan).
4) **Small object set** (3 YCB objects); generalization to a broader object distribution untested.
5) **Physics-level non-determinism** on marginal grasps ($\sim$ 0.6–1% outcome-flip rate on repeated identical execution) – a measurement-noise floor on any single-run success-rate estimate in this simulator, disclosed rather than hidden.
6) **Imitation-learning integration** (ACT) is a working pipeline, not a validated capability – first online rollout failed; 5 demonstrations/object is not enough data to expect closed-loop robustness. Reported as integration status, not as a result bearing on the paper's central claim.

VII. Toward Real-Hardware Validation

We designed, but did not execute, a protocol to test whether section V-B's simulated advantage transfers to physical SO-ARM101 hardware – a sim-to-real gap check, not a re-run of section V-B's statistical claims; a physical pilot that fails to replicate the simulated effect would not invalidate section V-B, and is the same class of outcome this project's own prior real-hardware work has already hit and reported honestly.

Object choice, corrected before any data collection. The critic's object one-hot only covers {CrackerBox, MustardBottle, PowerDrill} (section III-B); an earlier proposal to pilot on Pear was caught, during protocol design, as an out-of-distribution input the critic was never trained to handle – not a like-for-like generalization test. The protocol instead specifies CrackerBox and MustardBottle, both in-distribution and both objects this project already handles on real SO-ARM101 hardware. This correction is recorded here, before any trial, per this study's own no-post-hoc-selection discipline (section III-C).

Platform. SO-ARM101 with a relative-delta-clamped real backend, a RealSense D435i for pose estimation, and the same rotated-mount geometry and top-down IK bias already used in this study's simulation. Connectivity, calibration, and low-level motor control are already verified working from this project's independent real-hardware track; a thin real-hardware analogue of the simulation's candidate-pool-build/score/execute pipeline is new code to write, not new infrastructure to design.

Design. A matched-block paired trial (or, if placement restoration proves unreliable, an explicitly-declared unpaired two-proportion design instead – chosen before running, not after seeing results) with a small pre-registered pilot (12–20 total grasp attempts across both objects and methods) gating a larger scale-up (20–30 trials/object/condition): proceed only if the pilot's direction is consistent with simulation on both objects and no safety incident occurred; otherwise stop and report the sim-to-real gap honestly rather than scaling up in search of significance. A seven-item safety checklist (continuous human supervision, an active relative-motion clamp, pre-execution joint-range verification, workspace clearance, a confirmed e-stop path, reduced speed through the pilot stage, and an explicit collision check against the camera mount and tray) must be re-confirmed live at execution time, not assumed from this document.

What this would, and would not, settle. A completed pilot/scale-up would establish whether the simulated +14–16pp advantage survives the sim-to-real gap on CrackerBox and MustardBottle specifically, at the tested sample size. It would not establish generalization to PowerDrill or Pear, and does not test the risk gate at all (section VI) – only top-1 critic selection, to keep the physical pilot's scope minimal and interpretable.

Status, stated plainly. Camera repositioning is an unresolved physical-setup blocker, shared with an unrelated, independent line of work in this project, and is a precondition for every later stage. No real-hardware result from this section exists in this paper.

VIII. Conclusion

We set out to validate a pre-execution grasp critic and instead found that our own strongest reported result for it was invalid – not through a single bug, but two independent evaluation defects that each, alone, would have been enough to invalidate the comparison. Rather than treat that as a dead end, we applied and extended this project's causal-validity criterion to catch it, rebuilt the evaluation as a genuinely paired, causally-admissible design, and found a real, replicated, live-executed effect: an object-relative counterfactual critic beats a geometric baseline by +15.6pp on an independent development batch (p=0.00258, n=90) and +14.0pp on a frozen, pre-registered confirmatory batch never inspected before evaluation (p=3.24e-4, n=150), with a complementary offline decomposition explaining where and why the effect concentrates. We report this as the paper's central claim, and only this claim – not a claim about the specific pairwise loss term, which exact power analysis shows is not resolvable at the current data scale without roughly two orders of magnitude more evaluated candidate pools (section V-C), and not a claim about risk-aware gating, which measurably added nothing here (section VI). A real-hardware validation protocol is designed and the platform's connectivity and low-level control are independently verified, but no hardware result exists in this paper (section VII states the plan, not a result).

The discipline we believe generalizes beyond this specific pipeline is not the critic architecture or the loss function – it is the practice of auditing an evaluation's construction before trusting its output, reporting a negative gate result with the same rigor as a positive one, and stating plainly, at every step, what the evidence does and does not support.

Author Contributions

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Competing Interests

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Data Availability Statement

[FILL IN: state whether/where `pair_results.jsonl`, the trained critic checkpoints, and the evaluation harness will be made available, and under what conditions.]

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